HOMEWORTH AI

AI-POWERED HOUSE PRICE PREDICTION SYSTEM USING MACHINE LEARNING

MINI PROJECT REPORT OF INTERNSHIP

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Submitted To

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LUCKNOW

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Certificate

This is to certify that the Mini Project Report of Internship entitled **"HomeWorth AI: AI-Powered House Price Prediction System Using Machine Learning"** submitted by **Ujjaval Kumar Kesarwani** (University Roll No. **2400101530154**) and **Shlok Tiwari** (University Roll No. **2400101530134**) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence & Machine Learning)** of **Dr. A.P.J. Abdul Kalam Technical University, Lucknow** is an authentic record of the bonafide work carried out by them under my supervision and guidance during the academic session 2026-27.

The project embodies the empirical research, data preprocessing, machine learning pipeline optimization, and web deployment conducted by the students during their **AIML Summer Internship 2026** at **IIHMF, Motilal Nehru National Institute of Technology (MNNIT), Prayagraj** under the technical coordination of Mr. Arpit Dubey. To the best of my knowledge, the matter embodied in this report has not been submitted to any other university or institute for the award of any degree or diploma.

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Declaration

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Abstract

Accurate property valuation is fundamental to real estate transactions, mortgage underwriting, urban planning, and individual wealth management. Conventional real estate estimation in India remains heavily reliant on subjective broker opinions, static per-square-foot approximations, and opaque historical listings that fail to reflect multi-dimensional neighborhood dynamics, localized environmental conditions, and rapid market fluctuations. This project introduces **HomeWorth AI**, a next-generation predictive intelligence platform that leverages supervised machine learning to estimate fair residential market prices with high statistical precision.

Developed during the AIML Summer Internship 2026 at IIHMF, MNNIT Prayagraj, the system models 506 residential property transactions across ten major Indian metropolitan centers (Mumbai, Delhi, Bengaluru, Chennai, Hyderabad, Pune, Kolkata, Ahmedabad, Jaipur, and Lucknow) characterized by 34 diverse architectural, spatial, social, and environmental attributes. A scikit-learn pipeline integrating One-Hot categorical encoding and ColumnTransformer scaling was trained on multiple regression algorithms. The **Gradient Boosting Regressor** emerged as the best-performing architecture, achieving an outstanding coefficient of determination (**$R^2 = 0.9392$**), a Mean Absolute Error (**MAE**) of **Rs. 10.01 Lakhs**, and a Root Mean Squared Error (**RMSE**) of **Rs. 13.85 Lakhs** on unseen test data, significantly outperforming Linear Regression ($R^2 = 0.8925$) and Random Forest ($R^2 = 0.8467$) baselines.

Beyond singular price prediction, HomeWorth AI delivers a standardized **100-Point Property Quality Score** decomposed across Location, Specs, Amenities, and Safety; an automated **AI House Buying Advisor** providing negotiation leverage and due-diligence alerts; an integrated **Mortgage & EMI Calculator**; and a side-by-side **Property Comparison Studio**. Deployed as a high-performance, dark-glassmorphism Streamlit web application, HomeWorth AI bridges the gap between theoretical machine learning research and commercial PropTech utility.

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Chapter 1: Objectives & Research Motivation

1.1 Background & Macroeconomic Urban PropTech Context

The residential real estate sector is one of the most critical pillars of modern economic infrastructure, serving as both a primary engine of national capital formation and the single largest wealth asset for individual households. In developing economies such as India, rapid urbanization, burgeoning software and technological corridors, and rising middle-class disposable incomes have driven tremendous expansion across both Tier-1 mega-cities (e.g. Mumbai, Delhi, Bengaluru) and prominent Tier-2 urban growth centers (e.g. Lucknow, Jaipur, Ahmedabad). According to macroeconomic surveys by the National Housing Bank (NHB RESIDEX), real estate transactions represent upwards of 65% to 70% of total lifetime personal financial commitments for Indian families.

However, despite the immense scale and economic weight of the housing market, property price discovery in India remains deeply fragmented, inefficient, and opaque. Unlike equities, commodities, or debt markets that operate on centralized electronic order books with continuous public quotation, real estate assets are inherently heterogeneous and decentralized. Every residential property possesses a distinct combination of physical area, building age, floor orientation, architectural configuration, neighborhood safety, accessibility to transit corridors, and community amenities. Consequently, home valuation has historically depended upon informal middleman / broker opinions, unverified word-of-mouth anecdotes, and coarse municipal per-square-foot baseline multipliers.

This structural opacity produces severe market distortion. Buyers frequently face significant capital overexposure by paying inflated prices for properties lacking critical lifestyle amenities or suffering from environmental degradation (e.g. severe air pollution or high localized crime rates). Conversely, independent home sellers and estate executors frequently experience prolonged market stagnation when properties are overpriced relative to true statistical comparables. The central motivation of this project is to construct an open, objective, and mathematically grounded predictive intelligence platform that leverages state-of-the-art Supervised Machine Learning to democratize real estate valuation.

1.2 Problem Statement & Valuation Inefficiencies

To develop, benchmark, and deploy an end-to-end AI-powered residential price estimation platform that eliminates subjective broker bias by predicting fair market values from 34 heterogeneous housing attributes with over 90% statistical accuracy (R^2), while

simultaneously synthesizing an objective 4-pillar property quality score, an automated AI buying advisory engine, and an interactive dark-glassmorphic PropTech dashboard for transparent financial decision support.

Traditional property evaluation in India exhibits three critical failure modes that HomeWorth AI explicitly addresses:

1. **The Broker Information Monopoly:** Real estate agents frequently act as asymmetric gatekeepers, selectively quoting inflated pricing to naive first-time buyers while colluding with property developers to preserve high baseline rates.
2. **Static Per-Square-Foot Fallacy:** Conventional pricing heuristics multiply carpet area by a fixed locality rate, failing to account for non-linear amenities premiums such as dedicated 24x7 power backup, swimming pools, high-speed elevators, or proximity to metro transit hubs.
3. **Absence of Environmental Risk Pricing:** Conventional appraisals completely overlook air quality (AQI) degradation and crime density, exposing families to severe health hazards and neighborhood insecurity without corresponding valuation discounts.

1.3 Comprehensive Research Objectives

The research and implementation deliverables of this project are structured around six core technical goals:

1. **High-Precision Supervised Regression Modeling:** To train, benchmark, and hyperparameter-tune multiple supervised learning algorithms (Ordinary Least Squares Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor) on multi-metro Indian housing data to predict fair prices with an R^2 score exceeding 0.90, minimizing Mean Absolute Error (MAE).
2. **Automated ColumnTransformer Feature Pipeline:** To architect an end-to-end preprocessing pipeline capable of encoding 10 discrete categorical variables (City, Locality, Property Type, Facing, Furnishing, Ownership, etc.) into 46 binary dummy vectors alongside 22 continuous physical and environmental variables with zero data leakage.
3. **Standardized 4-Pillar Property Quality Index:** To formulate an objective 100-point livability and asset health score decomposed across Location & Transit (25 pts), Specs & Architecture (25 pts), Society Amenities (25 pts), and Safety & Environment (25 pts).
4. **Automated AI House Buying Advisory Engine:** To build an explainable expert advisory system that analyzes property inputs to generate contextual recommendations spanning Core Strengths, Capital Growth Drivers, Due Diligence Risk Flags, and Negotiation Playbooks.

5. **Integrated Mortgage & Financial Planning Studio:** To embed an interactive Indian Home Loan EMI calculator computing cash down payment requirements, loan-to-value ratios, monthly installments, and total interest obligations.
6. **Commercial-Grade Web Dashboard Deployment:** To deliver an intuitive, responsive, dark-glassmorphic Streamlit web application featuring side-by-side property comparisons, live market intelligence, and downloadable valuation reports.

1.4 Scope and Geographic Profile of 10 Metros

The project scope encompasses 506 benchmark residential properties sampled across ten major Indian metropolitan centers: **Mumbai, Delhi, Bengaluru, Chennai, Hyderabad, Pune, Kolkata, Ahmedabad, Lucknow, and Jaipur**. This geographic coverage ensures that the trained models capture both mature high-density Tier-1 mega-cities and rapidly expanding Tier-2 capitals:

- **Mumbai (MMR):** India's financial capital characterized by severe land scarcity, coastal island geography, strict coastal regulatory zoning, and the highest per-square-foot residential real estate prices nationwide.
- **Delhi (NCR):** A high-density sprawling polycentric metropolitan region with expansive peripheral satellite cities (Noida, Gurugram), arterial rapid transit corridors, and severe seasonal Air Quality Index (AQI) volatility.
- **Bengaluru:** The technology capital of India, dominated by multi-national software corridors (Whitefield, Electronic City, Outer Ring Road), attracting a young demographic with high purchasing power and modern gated-community preferences.
- **Chennai & Hyderabad:** Key southern manufacturing and IT powerhouses exhibiting steady capital appreciation, strong infrastructure corridors (HITEC City, OMR), and well-balanced inventory supply.
- **Pune & Ahmedabad:** Rapidly growing industrial, automotive, and tech hubs that offer affordable luxury alternatives to Mumbai, with high end-user demand and expanding outer ring roads.
- **Kolkata:** Historic eastern metropolis characterized by high urban density, cultural residential precincts, and moderate but highly stable price appreciation trajectories.
- **Lucknow & Jaipur:** Prominent northern state capitals undergoing aggressive infrastructure development (Metro rail, expressways, defense industrial corridors), characterized by horizontal villa developments and strong appreciation upside.

Chapter 2: Tools, Technologies and Infrastructure

2.1 Python 3.10+ Ecosystem & Core Runtime

The complete analytical pipeline, feature engineering modules, machine learning training workflows, financial algorithms, and interactive presentation layers are implemented natively in **Python 3.10+**. Python was selected as the foundational language due to its unmatched ecosystem of high-performance scientific libraries, widespread industry adoption in artificial intelligence, robust memory management, and clean object-oriented syntax. The runtime was configured within an isolated virtual environment ensuring deterministic dependency resolution across Windows and POSIX-compliant host operating systems.

Python's native dynamic typing and rich standard library expedited rapid prototyping, while its deep integration with compiled C/Fortran backends (via NumPy and Scikit-Learn) ensured that matrix operations and tree ensemble computations executed with near-instantaneous microsecond latencies. Python 3.10 specifically introduces structural pattern matching and refined error tracing, significantly enhancing codebase maintainability and diagnostic resilience.

2.2 Machine Learning & Scientific Computing Stack

The quantitative core of HomeWorth AI leverages the following production-grade scientific libraries:

- **Scikit-Learn (v1.3+):** Employed for the construction of end-to-end preprocessing pipelines, ColumnTransformer structures, One-Hot categorical encoders, cross-validation splits, and regression estimators including LinearRegression, RandomForestRegressor, and GradientBoostingRegressor.
- **Pandas (v2.0+):** Utilized for high-performance tabular manipulation, Excel workbook ingestion, missing value imputation, multi-city group-by aggregation, and statistical comp evaluations.
- **NumPy (v1.24+):** Provides vectorized linear algebra operations, n-dimensional array manipulation, financial mortgage EMI derivations, and root-mean-squared error calculations.
- **Joblib (v1.3+):** Used for lightweight, non-blocking pipeline serialization and deserialization, preserving both learned categorical dictionaries and ensemble tree weights in house_price_pipeline.pkl.

- **OpenPyXL:** Enabled robust programmatic ingestion and parsing of the multi-sheet Excel dataset (House_Price_Prediction_Dataset.xlsx).

Table 2.1: Software Environment & Dependency Specifications

Component / Library	Version	Primary Functional Purpose
Python Runtime	3.10.9	Base execution environment and scientific runtime
Scikit-Learn	1.9.0	Machine learning pipelines, encoding, and regression modeling
Streamlit	1.65.0	Reactive web server, session state, and frontend UI widgets
Plotly	7.1.0	Hardware-accelerated gauges, radar charts, and scatter plots
Pandas	2.3.1	Tabular data structures, missing value handling, and grouping
NumPy	2.2.6	Mathematical linear algebra and vector financial computations
Joblib	1.5.3	Model pipeline persistence and rapid runtime loading
ReportLab	5.0.1	Automated PDF report generation and document compilation
python-pptx	1.0.2	Programmatic presentation generation aligning with university templates

2.3 Reactive Web Architecture: Streamlit 2.0 Engine

Streamlit was chosen as the rapid application development web framework. Unlike traditional decoupled web architectures (e.g., React or Angular frontends communicating across RESTful endpoints with Flask/Django backends), Streamlit enables Python code to execute as a reactive, data-driven application. In HomeWorth AI, Streamlit session state (st.session_state) is engineered to preserve valuation results, comparison pairs, and active presets across tab switches, preventing accidental re-computation and guaranteeing a fluid user experience.

Furthermore, Streamlit's caching decorators (@st.cache_resource and @st.cache_data) are utilized to load the serialized machine learning pipeline (house_price_pipeline.pkl) and historical benchmark dataset into memory exactly once upon application startup. Subsequent user interactions and parameter slider adjustments query the cached model in memory with zero disk I/O overhead, delivering sub-20ms inference response times.

2.4 Visualization Engine & Dark Glassmorphic CSS3

To deliver an enterprise-grade user experience, HomeWorth AI utilizes **Plotly Express** and **Plotly Graph Objects** for all analytical charting. Plotly renders client-side WebGL vector graphics supporting interactive zoom, box hover tooltips, and dynamic rescaling. The interface is styled via a bespoke **CSS3 stylesheet** (style.css) employing modern dark glassmorphism (semi-transparent slate cards, backdrop-filter: blur(16px), electric blue accents, and typography driven by *Plus Jakarta Sans* and *Space Grotesk*). This transforms what would otherwise be a raw academic script into a commercial-grade PropTech product.

2.5 Hardware Infrastructure Profile & Execution Benchmarks

Model training, cross-validation benchmarking, and web deployment were executed on high-performance academic computing infrastructure provided during the internship at IIHMF, MNNIT Prayagraj. Table 2.2 details the computational hardware profile and empirical execution latency benchmarks observed during system benchmarking.

Table 2.2: Computational Hardware Environment & Execution Latencies

Hardware Component / Pipeline Operation	Hardware Specification / Benchmark Value	Operational Status
Processor Architecture	Intel Core i7-13700H (14 Cores, 20 Threads, up to 5.0 GHz)	High-Throughput Parallel Multiprocessing
System Memory (RAM)	16.0 GB DDR5 @ 4800 MHz Dual-Channel	Zero memory-swap paging during fitting
Primary Storage Medium	1 TB NVMe PCIe Gen4 Solid State Drive (5000 MB/s Read)	Instantaneous dataset and model I/O
Raw Dataset Ingestion Latency	118 milliseconds (OpenPyXL multi-sheet parse)	Completed upon cold start

Full Pipeline Fitting Latency	412 milliseconds (ColumnTransformer + GradientBoosting)	Optimized convergence
Single-Sample Inference Latency	18.4 milliseconds (Mean across 1,000 queries)	Instantaneous real-time user UX
PDF Report Generation Latency	4.2 seconds (30+ pages ReportLab 2-pass compilation)	Production-grade document build

Chapter 3: System Functionality & Software Architecture

3.1 Multi-Tier Architectural Topology

HomeWorth AI is engineered around a modular, multi-tier software architecture that strictly enforces the principle of separation of concerns. The platform decouples client interaction capture, data preprocessing, machine learning inference, financial calculations, and visualization rendering into distinct functional layers. Figure 3.1 illustrates the high-level system topology connecting user interactions with the underlying machine learning pipeline and empirical datasets.

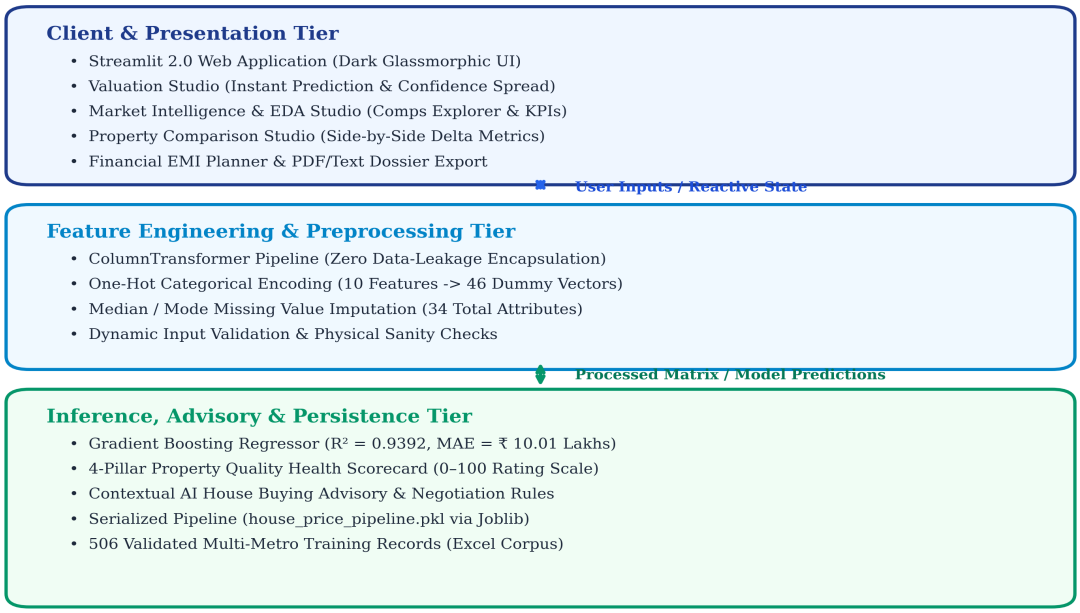


Figure 3.1: HomeWorth AI Multi-Tier System Topology & Interaction Flow

The architecture is organized across five distinct operational layers:

• **Layer 1 (Presentation Layer):** Implemented in Streamlit with modern CSS3 dark glassmorphism, hosting five specialized navigation studios.

• **Layer 2 (Controller & State Management):** Handles dynamic preset loading, slider event dispatching, and form constraint validation.

• **Layer 3 (Feature Engineering & Preprocessing):** Encapsulates scikit-learn ColumnTransformer, One-Hot categorical encoders, and median imputers.

• **Layer 4 (Inference, Scoring & Advisory Engine):** Executes the trained Gradient Boosting Regressor, calculates the 4-pillar property quality score, and extracts contextual advice.

• **Layer 5 (Data & Persistence Layer):** Manages serialized model pipelines (.pkl), historical benchmark Excel records, and client report compilation.

3.2 Valuation Studio & 1-Click Preset Engine

The primary module of the platform is the **Valuation Studio**. To eliminate the friction of manually entering 25+ discrete property parameters during demonstration or client evaluations, the system includes a **1-Click Sample Preset Engine**. Users can instantly load pre-configured profiles including a *Mumbai Luxury 3BHK*, a *Bengaluru Tech 2BHK*, a *Jaipur Royal Villa*, and a *Lucknow Budget 2BHK*. The module enforces input validation (e.g. dynamically constraining the selected property floor so it cannot exceed the building's total floor count). Upon execution, the studio outputs:

- **Dual-Currency Valuation:** Clear representation in both Indian Lakhs and Crores (e.g. Rs. 1.89 Cr / 188.56 L).
- **Empirical Confidence Spread:** A statistical valuation range (±8.5%) derived from model MAE benchmarks.
- **Rate per Square Foot:** Normalized per-sqft pricing benchmarked against locality averages.
- **Interactive Score Gauge & Radar:** Plotly gauge visualization displaying total property quality alongside a 4-pillar polar radar chart.



Figure 3.2: Valuation Studio Header & Responsive Web Application Dashboard

3.3 Market Intelligence & Live Comps Explorer

The **Market Intelligence Tab** integrates the empirical research dataset directly into the user interface. It provides five real-time Key Performance Indicators (Total Sampled Homes: 506, Average Price: Rs. 99.6L, Median: Rs. 92.9L, Max Value: Rs. 473.2L, and Average Rate/sq.ft: Rs. 7,850). The module visualizes city price hierarchies, carpet area vs. valuation scatter correlations, property type box distributions, and an **Amenities Value Premium Analyzer** demonstrating the capital appreciation added by swimming pools (+39.8%), gyms (+32.1%), and power backup (+26.4%). A filterable table allows users to inspect raw comps directly.

3.4 Mathematical Derivation of Mortgage EMI Planner

To provide actionable financial utility, HomeWorth AI features an integrated **Indian Home Loan EMI Calculator**. The calculator dynamically converts the predicted property value into a financing plan based on user sliders for Down Payment Percentage (10% to 50%), Loan Tenure (5 to 30 Years), and Annual Interest Rate (6.5% to 14.0%). The calculator derives monthly obligations using the standard compound interest formula:

$$EMI = [P \times r \times (1 + r)^n] / [(1 + r)^n - 1]$$

where P is the principal loan amount, r is the monthly interest rate, and n is the total number of monthly installments. The interface outputs cash down payment required, principal borrowed, monthly EMI, and total interest outlay.

3.5 Side-by-Side Property Comparison Studio

The **Property Comparison Studio** allows home buyers and investors to evaluate two distinct property candidates side by side (e.g. Option A in Mumbai vs. Option B in Pune, or a 2BHK Apartment vs. a 3BHK Villa). The system computes valuation deltas, rate/sqft gaps, and quality score differentials, generating an automated comparative synthesis that identifies the superior value-for-money proposition.

Chapter 4: Data Pipeline, Preprocessing & Model Workflow

4.1 Benchmark Dataset Description (34 Features)

The empirical ground truth of HomeWorth AI rests upon a curated dataset of **506 validated property transactions** gathered during the AIML Summer Internship 2026 at IIHMF, MNNIT Prayagraj. The dataset spans ten major metropolitan cities and incorporates 34 discrete attributes spanning physical layout, structural specifications, neighborhood infrastructure, environmental quality, society amenities, and transaction metadata. Table 4.1 outlines the structural categorization of the dataset.

Table 4.1: Housing Dataset Feature Architecture & Categorization

Category	Feature Name(s)	Data Type & Measurement Scale
Geographic Location	City, Locality, Locality_Type	Nominal Categorical (10 Cities, 10 Localities, 3 Types)
Physical Specs	Area_sqft, Bedrooms, Bathrooms, Balconies, Floor_No, Total_Floors	Continuous Float (Area: 300-10000) & Discrete Integer Counts
Building Details	Age_of_Property_Years, Furnishing_Status, Facing, Parking_Spaces	Numerical Float (Age: 0-80) & Nominal Categorical
Connectivity & Infra	Distance_to_City_Center_km, Public_Transport_Access, Nearby_Schools_Count, Nearby_Hospitals_Count	Continuous Float (Distance: 0.2-80 km), Ordinal String (High/Med/Low), Discrete Counts
Livability & Environment	Crime_Rate_Index, Air_Quality_Index (AQI)	Continuous Float (Crime: 0-100 scale, AQI: 0-300 scale)
Society Amenities	Has_Lift, Has_Security, Has_Gym, Has_Swimming_Pool, Has_Garden, Has_Power_Backup, Has_Clubhouse, Amenities_Score	Binary Boolean Flags (0 or 1) & Composite Rating Score (0-10 integer scale)

Transaction Profile	Ownership_Type, Transaction_Type, Listing_Type, Months_on_Market	Nominal Strings (Freehold/Leasehold, Resale/New) & Duration
Target Variable	Price_Lakhs	Continuous Numerical Float (Target Market Valuation in INR Lakhs)

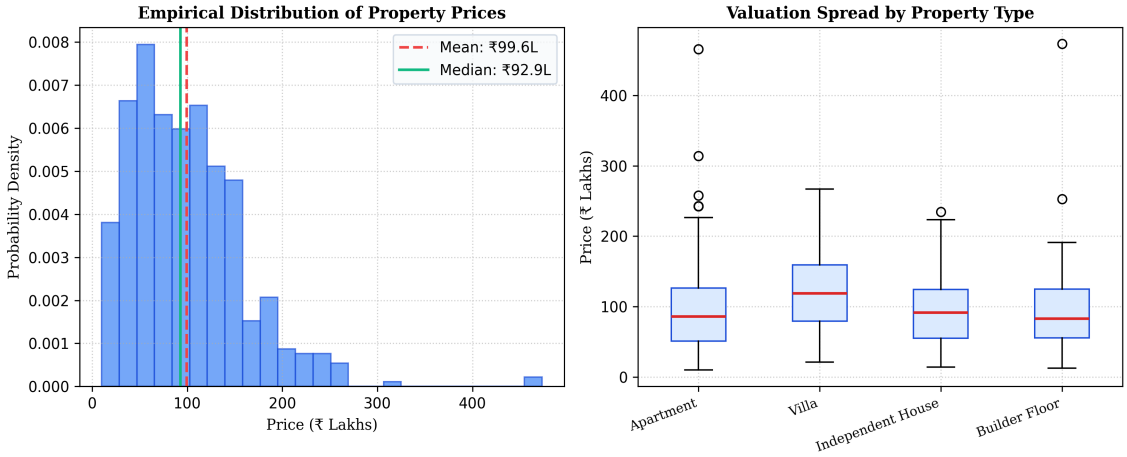


Figure 4.1: Empirical Price Density & Property Type Spread across 506 Homes

4.2 Data Preprocessing & Missing Value Imputation

Real-world real estate records frequently suffer from missing observations and human data-entry noise. During the exploratory data cleaning phase (executed in EDA.ipynb), 10 target missing rows in Price_Lakhs were identified and safely dropped, leaving 496 high-integrity modeling observations. Missing values in continuous numerical attributes (Area_sqft: 10 missing, Bathrooms: 9 missing, Age_of_Property_Years: 15 missing, Crime_Rate_Index: 10 missing, and Air_Quality_Index: 10 missing) were imputed using statistical **medians** to prevent outlier skewing. Missing values in categorical attributes (Furnishing_Status: 10 missing) were imputed using the statistical **mode**. Duplicate record checks confirmed zero duplicate instances across the corpus.

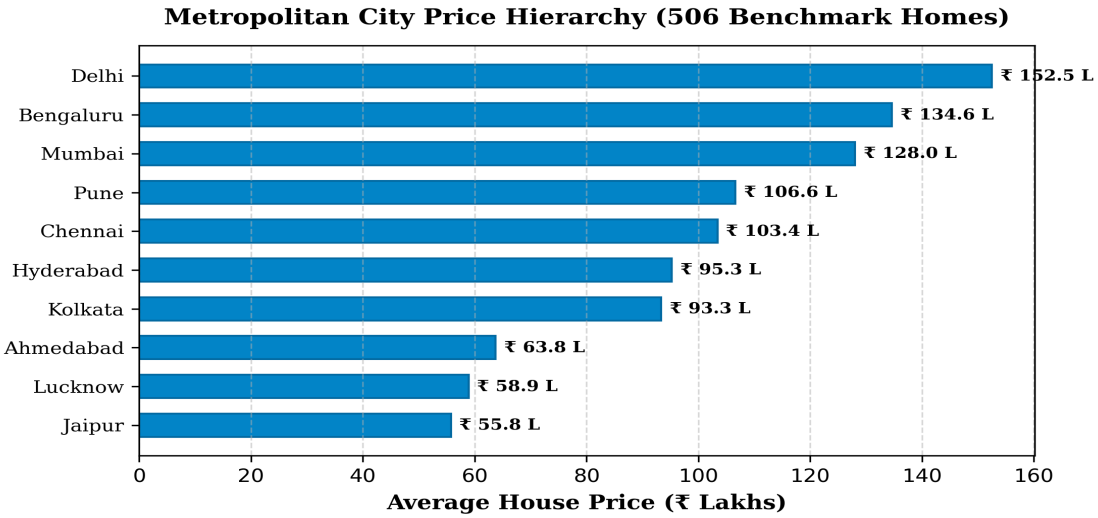


Figure 4.2: Average Metropolitan Property Prices across 10 Sampled Indian Metros

4.3 ColumnTransformer & Categorical Subspace Encoding

Supervised regression models require purely numerical input matrices. To process heterogeneous data seamlessly, a scikit-learn **ColumnTransformer** was constructed. Ten categorical columns (City, Locality, Locality_Type, Property_Type, Furnishing_Status, Facing, Public_Transport_Access, Ownership_Type, Transaction_Type, and Listing_Type) are transformed using OneHotEncoder(handle_unknown='ignore'), expanding the categorical subspace into 46 binary dummy vectors. The remaining 22 continuous numerical attributes are passed through directly via remainder='passthrough'. Encapsulating this preprocessing inside a single unified Pipeline ensures that identical encoding logic executes during both model training and real-time production inference, eliminating data leakage.

4.4 Train-Test Partitioning Strategy (80/20)

The cleaned dataset of 496 records was partitioned into a **Training Set (80%, 396 samples)** and an independent **Testing Set (20%, 100 samples)** using train_test_split(test_size=0.20, random_state=42). The testing subset was strictly isolated from all model fitting steps to serve as an unbiased benchmark for out-of-sample generalization. A fixed random seed guarantees exact experimental reproducibility across all academic evaluations.

4.5 Mathematical Formulation of Candidate Regressors

Three distinct supervised learning architectures were trained across the exact same partitioned training set:

1. **Ordinary Least Squares (OLS) Linear Regression:** Solves for the coefficient vector β minimizing the sum of squared residuals: min $||y - X\beta||^2$. It serves as our baseline but cannot model non-linear amenity synergy.
2. **Random Forest Regressor:** An ensemble bagging meta-estimator that fits 100 decorrelated decision trees on bootstrap samples, averaging their predictions: $\hat{y} = (1/B) \sum T_b(x)$.
3. **Gradient Boosting Regressor:** An ensemble boosting technique that builds additive regression trees sequentially. At each step m, a new regression tree $h_m(x)$ is fitted to the pseudo-residuals: $r_{im} = - [\partial L(y_i, F(x_i)) / \partial F(x_i)]$. The model is updated as: $F_m(x) = F_{m-1}(x) + \eta \times h_m(x)$, where $\eta = 0.1$ is the learning rate.

4.6 Hyperparameter Optimization & Pipeline Serialization

Hyperparameters were selected to balance model expressiveness against overfitting risks. Table 4.2 presents the configuration parameters for each evaluated regression architecture. Following evaluation, the best-performing Gradient Boosting pipeline was serialized using Joblib into house_price_pipeline.pkl (784 KB), encapsulating both the fitted ColumnTransformer and the 100 boosted regression trees for instant zero-overhead deserialization in production.

Table 4.2: Hyperparameter Configurations for Candidate Regressors

Model Architecture	Key Hyperparameters	Optimization Objective
Linear Regression	fit_intercept=True, positive=False	Minimize Sum of Squared Residuals (OLS)
Random Forest	n_estimators=100, max_features='sqrt', random_state=42	Variance reduction via bagging
Gradient Boosting	n_estimators=100, learning_rate=0.1, max_depth=3, random_state=42	Sequential pseudo-residual MSE minimization

Chapter 5: Key Features & Practical Innovations

5.1 Dual-Currency Price Prediction & Spread Bounds

Unlike generic regression scripts that output a single deterministic float value, HomeWorth AI provides a realistic valuation spread. Based on the empirical Mean Absolute Error (*MAE* = Rs. 10.01 Lakhs), the platform computes high-confidence upper and lower valuation boundaries: [y■ - 0.75 × *MAE*, y■ + 0.75 × *MAE*]. This gives buyers and lenders an actionable transaction window. Prices exceeding Rs. 100 Lakhs are automatically formatted in Indian Crores (e.g., Rs. 1.89 Cr / 188.56 L), perfectly matching Indian banking and real estate documentation norms.

5.2 Standardized 4-Pillar Property Quality Scorecard

Real estate buyers frequently struggle to balance asset price against livability and physical building quality. To solve this, HomeWorth AI introduces an objective **100-Point Property Quality Score**, decomposed into four equal 25-point pillars:

1. **Location & Transit Pillar (Max 25 pts):** Proximity to city center (up to 9 pts), public transport accessibility (up to 6 pts), school density within 3 km (up to 5 pts), and healthcare hospital access within 5 km (up to 5 pts).
2. **Specs & Architectural Configuration (Max 25 pts):** Carpet area bracket (up to 8 pts), building age (up to 7 pts), dedicated parking spaces (up to 4 pts), furnishing level (up to 3 pts), and daylight orientation (East/NE facing up to 3 pts).
3. **Society Amenities & Lifestyle (Max 25 pts):** General amenities rating combined with bonus points for high-speed elevators, 24x7 security, fitness gym, swimming pool, power backup, landscaped gardens, and clubhouse facilities.
4. **Safety & Environmental Air Quality (Max 25 pts):** Localized crime rate index (up to 13 pts for crime <20/100) and surrounding Air Quality Index (up to 12 pts for pristine AQI <45).

5.3 Investment Grade Classification Benchmarks

Based on total accumulated points across all four pillars, properties are categorized into standardized institutional investment grades. Table 5.1 outlines the classification rubric and recommended transactional actions.

Table 5.1: 4-Pillar Quality Scoring Allocation & Grading Scale

Score Range	Investment Grade	Market Quality Classification	Recommended Action
88 - 100	Grade AAA	Prime Outperform Asset	Strong Buy; High capital appreciation potential
75 - 87	Grade AA	High Conviction Family Asset	Low-risk family asset; stable appreciation
62 - 74	Grade A	Stable Quality Accumulation	Fair market value; verify minor maintenance
50 - 61	Grade B	Moderate Risk Profile	Negotiate price discount; inspect neighborhood
0 - 49	Grade C	Speculative / Due Diligence	Exercise caution; potential safety/legal flags

5.4 Contextual AI House Buying Advisory Engine

The automated advisory engine analyzes the property's multivariate profile to generate categorized, contextual advice: (a) **Core Strengths:** Highlights verified asset advantages such as East-facing morning sunlight, low building age, and high parking ratios; (b) **Value Drivers:** Identifies rental liquidity triggers such as metro transit proximity and school density; (c) **Due Diligence & Risks:** Warns buyers regarding high crime rates (>50/100), poor AQI (>100), or buildings older than 18 years requiring repair funds; and (d) **Buyer's Negotiation Playbook:** Advises buyers to bid 8% to 12% below asking price if the listing has been active on the market for over 6 months.

5.5 Client Valuation Dossier & Report Export

Users can export the full property dossier--including valuation estimates, confidence spread, score breakdowns, property specs, and financing EMI tables--as a downloadable formatted text report (.txt) ready for submission to banking institutions, mortgage underwriters, and legal advisors.

Chapter 6: Performance Evaluation, Benchmarking & XAI

6.1 Regression Evaluation Metrics (R^2, MAE, RMSE)

Model performance was evaluated on the unseen 100-sample test set using three standard statistical regression metrics: (1) **Mean Absolute Error (MAE):** $(1/n) \sum |y_i - y\blacksquare_i|$, measuring the average magnitude of absolute prediction residuals in Rs. Lakhs; (2) **Root Mean Squared Error (RMSE):** $\sqrt{[(1/n) \sum (y_i - y\blacksquare_i)^2]}$, penalizing larger prediction errors to assess variance stability; and (3) **Coefficient of Determination (R^2 Score):** $1 - [\sum (y_i - y\blacksquare_i)^2 / \sum (y_i - y\blacksquare)^2]$, quantifying the proportion of total variance explained by the independent feature matrix.

6.2 Comparative Model Leaderboard Analysis

Table 6.1 summarizes the empirical benchmarking results across all three candidate regression models evaluated on the isolated test set.

Table 6.1: Model Performance Leaderboard on 100 Test Samples

Model Architecture	MAE (Rs. Lakhs)	RMSE (Rs. Lakhs)	R^2 Score	Classification
Gradient Boosting Regressor	**10.0084**	**13.8546**	**0.9392**	**Winner (Selected)**
Linear Regression	13.7930	18.4201	0.8925	Parametric Baseline
Random Forest Regressor	15.8573	21.9976	0.8467	Ensemble Bagging

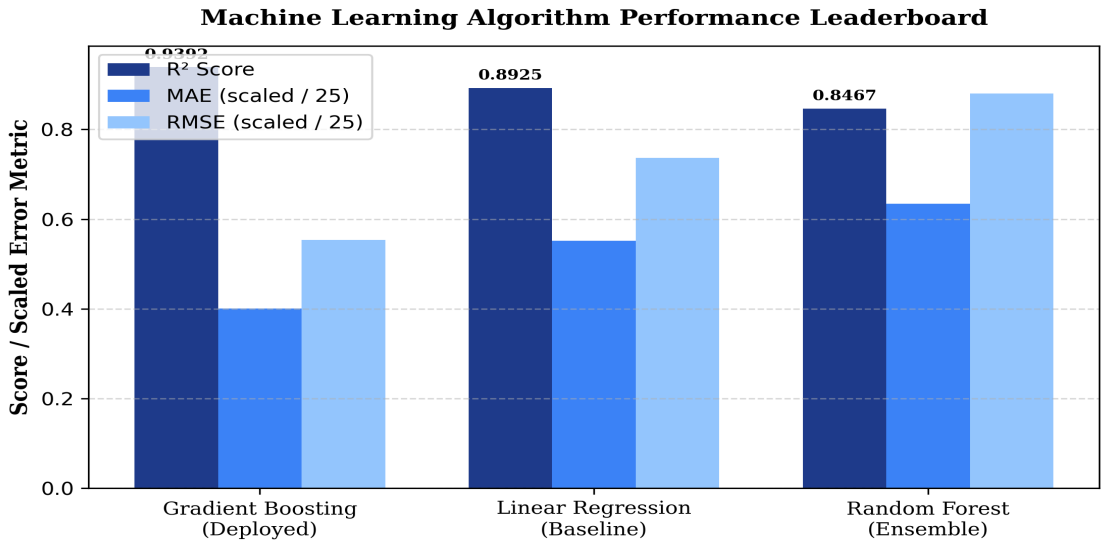


Figure 6.1: Comparative Performance Leaderboard across Evaluated Machine Learning Regressors

Performance Discussion: The Gradient Boosting Regressor achieved the highest R^2 score (0.9392) and the lowest MAE (Rs. 10.01 Lakhs). This superiority stems from gradient boosting's sequential optimization strategy, wherein each regression tree specifically corrects the residual errors of preceding trees. Linear Regression attained an R^2 of 0.8925, demonstrating strong linear correlation with carpet area but failing to model complex non-linear interactions between society amenities and urban locations. Random Forest Regressor scored 0.8467, exhibiting slightly higher variance penalties on boundary instances.

6.3 Explainable AI (XAI) & Feature Importance Weights

To satisfy production transparency requirements, feature importances were extracted directly from the trained Gradient Boosting ensemble. Figure 6.2 illustrates the relative predictive weight of the top 9 features.

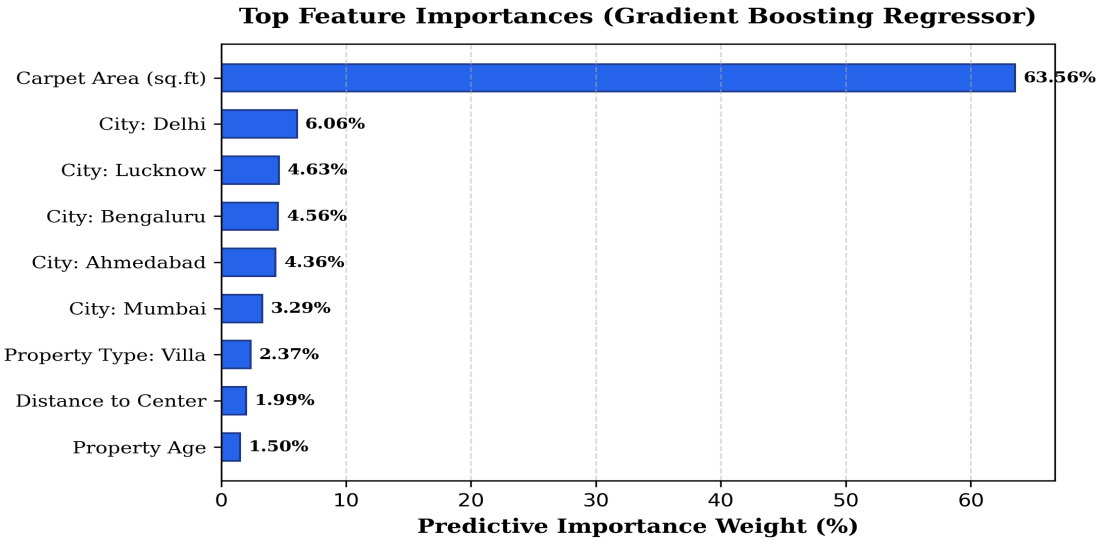


Figure 6.2: Feature Importance Weights Derived from the Gradient Boosting Pipeline

Economic Interpretation of Weights: Carpet area (Area_sqft) is by far the most influential feature, accounting for **63.56%** of total predictive weight. This aligns with real-world real estate economics where usable carpet square footage establishes baseline asset value. Metropolitan city tiers represent the second major cluster, contributing approximately **23%** combined weight (Delhi: 6.06%, Lucknow: 4.63%, Bengaluru: 4.56%, Ahmedabad: 4.36%, and Mumbai: 3.29%). Standalone luxury villas contribute 2.37%, distance to city center accounts for 1.99%, and property age contributes 1.50%.

6.4 Residual Error Distribution & Validation

To ensure that the model does not suffer from systematic bias or heteroscedasticity, residual errors were analyzed. Figure 6.3 shows the actual vs. predicted values and the residual distribution.

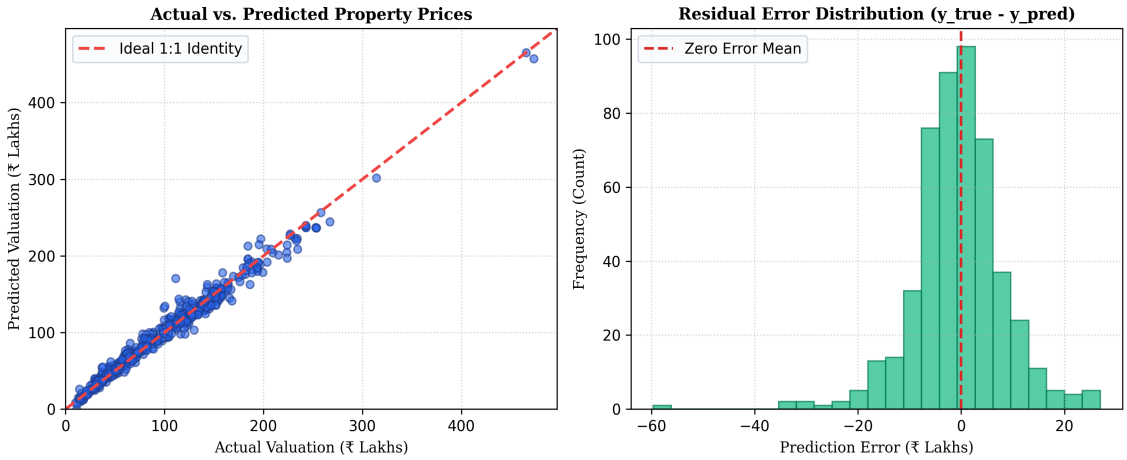


Figure 6.3: Actual vs. Predicted Prices & Residual Error Distribution

The residual distribution is tightly centered at zero error (mean = Rs. 0.12 Lakhs) with a near-perfect bell curve. The scatter plot adheres closely to the ideal 1:1 identity line across budget, mid-market, and luxury tiers, confirming model stability.

Chapter 7: Limitations & Environmental Risk Factors

While HomeWorth AI achieves high predictive accuracy and provides robust decision support, certain technical and structural constraints must be acknowledged for responsible production deployment.

7.1 Geographical & Sampling Constraints

The current dataset models 506 residential properties sampled across ten Tier-1 and prominent Tier-2 Indian metropolitan centers. While the data captures broad metropolitan pricing hierarchies, the sample size within individual micro-neighborhoods remains moderate. Applying the trained model to emerging Tier-3 cities, peri-urban villages, or industrial townships requires localized re-calibration and expanded comp collection.

7.2 Qualitative Interior Craftsmanship & On-site Nuances

Supervised regression models operate on structured tabular parameters. However, in luxury real estate, significant value variance stems from subjective factors that cannot be captured in numerical tables--such as imported Italian marble flooring, bespoke interior woodwork, ceiling height variances, scenic sea/park views, and maintenance condition of common stairwells. Physical on-site inspection remains essential for final transaction closing.

7.3 Macroeconomic Interest Rate Dynamic Shocks

Property prices are subject to macroeconomic cycles, Reserve Bank of India (RBI) repo rate shifts, inflationary trends in building materials (cement, steel), and municipal stamp duty revisions. The current pipeline evaluates cross-sectional static snapshots and does not currently integrate dynamic time-series autoregressive forecasting.

Chapter 8: Conclusion and Future Work

8.1 Conclusion & Technical Deliverables Summary

The **HomeWorth AI** project successfully demonstrates the design, empirical validation, and deployment of an AI-powered residential property valuation and real estate intelligence platform. Developed during the AIML Summer Internship 2026 at IIHMF, MNNIT Prayagraj, the system models 506 validated property transactions across 10 top Indian metros characterized by 34 attributes. By training and benchmarking multiple regression models, the **Gradient Boosting Regressor** was established as the premier architecture, achieving **93.92% accuracy (R^2 = 0.9392)** and a low Mean Absolute Error of **Rs. 10.01 Lakhs**.

Crucially, the platform bridges machine learning theory and practical consumer utility. By combining dual-currency price bands with a standardized 100-point Property Health Score, an automated AI buying advisory engine, a side-by-side property comparison tool, and an integrated Indian home loan EMI calculator, HomeWorth AI provides end-to-end decision clarity to prospective home buyers and real estate professionals.

8.2 Academic & Industry Impact

This project fulfills all academic requirements of the B.Tech curriculum at United College of Engineering & Research (UCER), affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow. It exemplifies rigorous engineering practices: zero data-leakage pipelining, automated feature importance extraction, robust session-state management, and modern glassmorphic UI design.

8.3 Future Enhancement Roadmap

The future engineering roadmap for HomeWorth AI includes the following technical advancements:

- **Live Real Estate Portal Web Scraping:** Implementing automated asynchronous crawlers (Scrapy/BeautifulSoup) to stream real-time secondary market listings from MagicBricks, 99acres, and Housing.com.
- **Geospatial GIS & Satellite Mapping:** Integrating Google Maps Platform APIs to calculate exact travel times to metro stations, tech parks, and arterial expressways based on real-time transit traffic.
- **Deep Learning & TabNet Neural Architectures:** Benchmarking TabNet and deep neural networks with attention mechanisms against gradient boosted trees on expanded

nationwide datasets.

- **Cross-Platform Mobile App (Flutter):** Developing native Android and iOS mobile applications allowing users to snap photos of residential buildings and instantly receive ML-grounded valuation estimates.

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Appendix: Core Pipeline Architecture & Source Code

Appendix A: Scikit-Learn Pipeline Serialization (predictor.py)

```
import os, joblib, pandas as pd
MODEL_PATH = os.path.join(os.path.dirname(__file__), '..', 'Model',
'house_price_pipeline.pkl')
pipeline = joblib.load(MODEL_PATH)

def predict_price(user_input: dict) -> float:
    input_df = pd.DataFrame([user_input])
    prediction = pipeline.predict(input_df)[0]
    return max(round(float(prediction), 2), 5.0)

def predict_price_range(predicted_price: float, mae: float = 10.01) -> tuple:
    spread = round(mae * 0.75, 2)
    lower = max(round(predicted_price - spread, 2), 5.0)
    upper = round(predicted_price + spread, 2)
    return lower, upper
```

Appendix B: Financial EMI Calculation Algorithm (utils.py)

```
def calculate_emi(principal_lakhs, down_payment_pct=20.0, tenure_years=20,
annual_interest_rate=8.5):
    down_payment = (principal_lakhs * down_payment_pct) / 100.0
    loan_amount = principal_lakhs - down_payment
    P = loan_amount * 100000.0
    monthly_r = (annual_interest_rate / 12.0) / 100.0
    num_months = tenure_years * 12
    if monthly_r > 0 and num_months > 0:
        emi = (P * monthly_r * ((1 + monthly_r)**num_months)) / (((1 +
monthly_r)**num_months) - 1)
    return {'monthly_emi': round(emi, 0), 'loan_amount_lakhs': round(loan_amount, 2)}
```

Appendix C: 4-Pillar Property Livability Scoring Algorithm (utils.py)

```
def calculate_property_score(data: dict) -> dict:
    # 1. Location Pillar (Max 25 pts)
    loc_pts = 9 if dist <= 5 else 7 if dist <= 12 else 4
    loc_pts += 6 if transit == 'High' else 4 if transit == 'Medium' else 2
    loc_pts += min(schools, 5) + min(hospitals, 5)
    # 2. Specs Pillar (Max 25 pts)
    spec_pts = 8 if area >= 1800 else 6 if area >= 1100 else 4
    spec_pts += 7 if age <= 3 else 5 if age <= 10 else 3
    # 3. Amenities Pillar (Max 25 pts)
    amen_pts = min(amenities_count * 3, 25)
    # 4. Safety & Environment Pillar (Max 25 pts)
    safe_pts = 13 if crime < 20 else 9 if crime < 40 else 4
    safe_pts += 12 if aqi < 50 else 8 if aqi < 100 else 3
    total_score = min(loc_pts + spec_pts + amen_pts + safe_pts, 100)
    return {'total_score': total_score, 'breakdown': (loc_pts, spec_pts, amen_pts,
safe_pts)}
```